u_alu

alu_32bit

CF n/c OF n/c SF n/c ZF
a[31:0] alu_ctrl[2:0] b[31:0] result[31:0] u_adder

adder_32bit

CFOF

SF

ZF n/c

sub

a[31:0] b[31:0]

result[31:0]

u_high

cla_16bit

cin

cout gg n/c pg n/c a[15:0] b[15:0]

sum[15:0]

u_low

cla_16bit

cin

cout gg n/c pg n/c a[15:0] b[15:0]

sum[15:0]

u0

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u1

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u2

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u3

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u0

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u1

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u2

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u3

cla_4bit

cin

cout n/c gg

pg

a[3:0] b[3:0] sum[3:0]

u_control

control_unit

alu_src branch mem_to_reg mem_write

reg_write

alu_ctrl[2:0]

funct3[2:0]

funct7[6:0] opcode[6:0]

u_dmem

data_memory

clk we

addr[31:0] rdata[31:0] wdata[31:0] u_imem

instruction_memory

addr[31:0] instr[31:0] u_imm

imm_extend

imm_ext[31:0]

instr[31:0]

u_regfile

rv32_regfile

clk

we

raddr1[4:0]

raddr2[4:0]

rdata1[31:0]

rdata2[31:0]

waddr[4:0] wdata[31:0]

OF_i RTL_OR

I0

I1 O OF0_i RTL_AND

I0

I1 O OF0_i__0 RTL_AND

I0

I1 O OF1_i RTL_AND

I0

I1 O OF1_i__0 RTL_AND

I0

I1 O =ZF_i RTL_EQO

I0[31:0]

I1[31:0]

b_eff_i RTL_XOR

I0[31:0]

I1[31:0] O[31:0]

regs_reg[30]_i RTL_ROM

A[4:0] O

c4_i RTL_OR

I0

I1 O

c8_i RTL_OR

I0

I1 O
c12_i RTL_OR

I0

I1 O c16_i RTL_OR

I0

I1 O

c40_i RTL_AND

I0

I1 O

c80_i RTL_OR

I0

I1 O

c80_i__0 RTL_AND

I0

I1 O

c81_i RTL_AND

I0

I1 O

c81_i__0 RTL_AND

I0

I1 O

c120_i RTL_OR

I0

I1 O

c120_i__0 RTL_AND

I0

I1 O

c121_i RTL_OR

I0

I1 O

c121_i__0 RTL_AND

I0

I1 O

c121_i__1 RTL_AND

I0

I1 O

c122_i RTL_AND

I0

I1 O

c122_i__0 RTL_AND

I0

I1 O

c160_i RTL_AND

I0

I1 O

c161_i RTL_AND

I0

I1 O

gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O

pg_i RTL_REDUCTION_AND

I0[3:0] O

regs_reg[30][31:0] RTL_REG C

CED Q

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

regs_reg[29]_i RTL_ROM

A[4:0] O

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

regs_reg[29][31:0] RTL_REG C

CED Q

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

regs_reg[28]_i RTL_ROM

A[4:0] O

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

regs_reg[28][31:0] RTL_REG C

CED Q

c4_i RTL_OR

I0

I1 O

c8_i RTL_OR

I0

I1 O
c12_i RTL_OR

I0

I1 O c16_i RTL_OR

I0

I1 O

c40_i RTL_AND

I0

I1 O

c80_i RTL_OR

I0

I1 O

c80_i__0 RTL_AND

I0

I1 O

c81_i RTL_AND

I0

I1 O

c81_i__0 RTL_AND

I0

I1 O

c120_i RTL_OR

I0

I1 O

c120_i__0 RTL_AND

I0

I1 O

c121_i RTL_OR

I0

I1 O

c121_i__0 RTL_AND

I0

I1 O

c121_i__1 RTL_AND

I0

I1 O

c122_i RTL_AND

I0

I1 O

c122_i__0 RTL_AND

I0

I1 O

c160_i RTL_AND

I0

I1 O

c161_i RTL_AND

I0

I1 O

gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O

pg_i RTL_REDUCTION_AND

I0[3:0] O

regs_reg[27]_i RTL_ROM

A[4:0] O

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

regs_reg[27][31:0] RTL_REG C

CED Q

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

regs_reg[26]_i RTL_ROM

A[4:0] O

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

regs_reg[26][31:0] RTL_REG C

CED Q

c1_i RTL_OR

I0

I1 O

c2_i RTL_OR

I0

I1 O

c3_i RTL_OR

I0

I1 O

c4_i RTL_OR

I0

I1 O

c10_i RTL_AND

I0

I1 O

c20_i RTL_OR

I0

I1 O

c20_i__0 RTL_AND

I0

I1 O

c21_i RTL_AND

I0

I1 O

c21_i__0 RTL_AND

I0

I1 O

c30_i RTL_OR

I0

I1 O

c30_i__0 RTL_AND

I0

I1 O

c31_i RTL_OR

I0

I1 O

c31_i__0 RTL_AND

I0

I1 O

c31_i__1 RTL_AND

I0

I1 O

c32_i RTL_AND

I0

I1 O

c32_i__0 RTL_AND

I0

I1 O

c40_i RTL_AND

I0

I1 O c41_i RTL_AND

I0

I1 O

g_i RTL_AND

I0[3:0]

I1[3:0] O[3:0] gg_i RTL_OR

I0

I1 O

gg0_i RTL_OR

I0

I1 O

gg0_i__0 RTL_AND

I0

I1 O

gg1_i RTL_OR

I0

I1 O

gg1_i__0 RTL_AND

I0

I1 O

gg1_i__1 RTL_AND

I0

I1 O

gg2_i RTL_AND

I0

I1 O

gg2_i__0 RTL_AND

I0

I1 O p_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0] pg_i RTL_REDUCTION_AND

I0[3:0] O

sum_i RTL_XOR

I0[3:0]

I1[3:0] O[3:0]

use_sub_i RTL_OR

I0

I1 O

use_sub0_i RTL_OR

I0

I1 O =

use_sub0_i__0 RTL_EQO

I0[2:0] V=B"100" I1[2:0] =

use_sub1_i RTL_EQO

I0[2:0] V=B"001" I1[2:0] =

use_sub1_i__0 RTL_EQO

I0[2:0] V=B"011" I1[2:0]

regs_reg[25]_i RTL_ROM

A[4:0] O

alu_ctrl_i RTL_MUX

S=3'b000 I0[1:0] V=B"010", S=3'b110 I1[2:0] V=B"011", S=3'b010 I2[2:0] V=B"100", S=3'b011 I3[2:0] S=default I4[2:0] O[2:0] S[2:0] alu_ctrl_i__0 RTL_MUX

S=7'b0110011 I0[2:0] S=7'b0000011 I1[2:0] S=7'b0100011 I2[2:0] V=B"001", S=7'b1100011 I3[2:0] V=B"101", S=7'b0110111 I4[2:0] S=default I5[2:0] O[2:0] S[6:0]

alu_ctrl0_i RTL_MUX S

V=B"01", S=1'b1 I0[1:0] S=default I1[1:0] O[1:0]

alu_src_i RTL_ROM

A[6:0] O

branch_i RTL_ROM

A[6:0] O mem_to_reg_i RTL_ROM
A[6:0] O mem_write_i RTL_ROM

A[6:0] O

reg_write_i RTL_ROM

A[2:0] O

reg_write_i__0 RTL_MUX

S=7'b0110011 I0

S=7'b0000011 I1

S=7'b0100011 I2

S=7'b1100011 I3

S=7'b0110111 I4

S=default I5 O
S[6:0]

regs_reg[25][31:0] RTL_REG C

CED Q

ram_reg RTL_RAM

WCLK WE1 RA1[7:0] RO1[31:0] WA1[7:0] WD1[31:0]

regs_reg[24]_i RTL_ROM

A[4:0] O

instr_i RTL_ROM

A[7:0] O[31:0]

regs_reg[24][31:0] RTL_REG C

CED Q

imm_ext_i RTL_MUX

S=7'b0000011 I0[31:0] S=7'b0100011 I1[31:0] S=7'b1100011 I2[31:0] S=7'b0110111 I3[31:0] S=default I4[31:0] O[31:0] S[6:0]

regs_reg[23]_i RTL_ROM

A[4:0] O

RTL_AND

RTL_AND

I0

I1 O

!=RTL_NEQ
RTL_NEQ O

I0[4:0]

I1[4:0]

rdata1_i RTL_MUX

S=5'b00000 I0[31:0] S=default I1[31:0] O[31:0] S[4:0]

rdata2_i RTL_MUX

S=5'b00000 I0[31:0] S=default I1[31:0] O[31:0] S[4:0]

regs_i RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__0 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__2 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__3 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__4 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__5 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__6 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__7 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__8 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__9 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__10 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__11 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__12 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__13 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__14 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__15 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__16 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__17 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__18 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__19 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__20 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__21 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__22 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__23 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__24 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__25 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__26 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__27 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__28 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__29 RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_i__30

RTL_MUX

S=5'b00000 I0[31:0] S=5'b00001 I1[31:0] S=5'b01010 I10[31:0] S=5'b01011 I11[31:0] S=5'b01100 I12[31:0] S=5'b01101 I13[31:0] S=5'b01110 I14[31:0] S=5'b01111 I15[31:0] S=5'b10000 I16[31:0] S=5'b10001 I17[31:0] S=5'b10010 I18[31:0] S=5'b10011 I19[31:0] S=5'b00010 I2[31:0] S=5'b10100 I20[31:0] S=5'b10101 I21[31:0] S=5'b10110 I22[31:0] S=5'b10111 I23[31:0] S=5'b11000 I24[31:0] S=5'b11001 I25[31:0] S=5'b11010 I26[31:0] S=5'b11011 I27[31:0] S=5'b11100 I28[31:0] S=5'b11101 I29[31:0] S=5'b00011 I3[31:0] S=5'b11110 I30[31:0] S=5'b11111 I31[31:0] S=5'b00100 I4[31:0] S=5'b00101 I5[31:0] S=5'b00110 I6[31:0] S=5'b00111 I7[31:0] S=5'b01000 I8[31:0] S=5'b01001 I9[31:0]

O[31:0]

S[4:0] regs_i__31

RTL_MUX

S=5'b00000 I0[31:0] S=5'b00001 I1[31:0] S=5'b01010 I10[31:0] S=5'b01011 I11[31:0] S=5'b01100 I12[31:0] S=5'b01101 I13[31:0] S=5'b01110 I14[31:0] S=5'b01111 I15[31:0] S=5'b10000 I16[31:0] S=5'b10001 I17[31:0] S=5'b10010 I18[31:0] S=5'b10011 I19[31:0] S=5'b00010 I2[31:0] S=5'b10100 I20[31:0] S=5'b10101 I21[31:0] S=5'b10110 I22[31:0] S=5'b10111 I23[31:0] S=5'b11000 I24[31:0] S=5'b11001 I25[31:0] S=5'b11010 I26[31:0] S=5'b11011 I27[31:0] S=5'b11100 I28[31:0] S=5'b11101 I29[31:0] S=5'b00011 I3[31:0] S=5'b11110 I30[31:0] S=5'b11111 I31[31:0] S=5'b00100 I4[31:0] S=5'b00101 I5[31:0] S=5'b00110 I6[31:0] S=5'b00111 I7[31:0] S=5'b01000 I8[31:0] S=5'b01001 I9[31:0]

O[31:0]

S[4:0]

regs_reg[0][31:0] RTL_REG CD Q

regs_reg[1][31:0] RTL_REG C

CED

regs_reg[1]_i Q RTL_ROM

A[4:0] O

regs_reg[2][31:0] RTL_REG C

CED Q
regs_reg[2]_i RTL_ROM

A[4:0] O

regs_reg[3][31:0] RTL_REG C

CED Q

regs_reg[3]_i RTL_ROM

A[4:0] O

regs_reg[4][31:0] RTL_REG C

CED Q

regs_reg[4]_i RTL_ROM

A[4:0] O

regs_reg[5][31:0] RTL_REG C

CED Q

regs_reg[5]_i RTL_ROM

A[4:0] O

regs_reg[6][31:0] RTL_REG C

CED Q

regs_reg[6]_i RTL_ROM

A[4:0] O

regs_reg[7][31:0] RTL_REG C

CED Q

regs_reg[7]_i RTL_ROM

A[4:0] O

regs_reg[8][31:0] RTL_REG C

CED Q

regs_reg[8]_i RTL_ROM

A[4:0] O

regs_reg[9][31:0] RTL_REG C

CED Q

regs_reg[9]_i RTL_ROM

A[4:0] O

regs_reg[10][31:0] RTL_REG C

CED Q

regs_reg[10]_i RTL_ROM

A[4:0] O

regs_reg[11][31:0] RTL_REG C

CED Q

regs_reg[11]_i RTL_ROM

A[4:0] O

regs_reg[12][31:0] RTL_REG C

CED Q

regs_reg[12]_i RTL_ROM

A[4:0] O

regs_reg[13][31:0] RTL_REG C

CED Q

regs_reg[13]_i RTL_ROM

A[4:0] O

regs_reg[14][31:0] RTL_REG C

CED Q

regs_reg[14]_i RTL_ROM

A[4:0] O

regs_reg[15][31:0] RTL_REG C

CED Q

regs_reg[15]_i RTL_ROM

A[4:0] O

regs_reg[16][31:0] RTL_REG C

CED Q

regs_reg[16]_i RTL_ROM

A[4:0] O

regs_reg[17][31:0] RTL_REG C

CED Q

regs_reg[17]_i RTL_ROM

A[4:0] O

regs_reg[18][31:0] RTL_REG C

CED Q

regs_reg[18]_i RTL_ROM

A[4:0] O

regs_reg[19][31:0] RTL_REG C

CED Q

regs_reg[19]_i RTL_ROM

A[4:0] O

regs_reg[20][31:0] RTL_REG C

CED Q

regs_reg[20]_i RTL_ROM

A[4:0] O

regs_reg[21][31:0] RTL_REG C

CED Q

regs_reg[21]_i RTL_ROM

A[4:0] O

regs_reg[22][31:0] RTL_REG C

CED Q

regs_reg[22]_i RTL_ROM

A[4:0] O

regs_reg[23][31:0] RTL_REG C

CED Q

CF0_i RTL_INV

I0 O CF_i RTL_MUX

S=1'b1 I0

S=default I1 O S

regs_reg[31][31:0] RTL_REG C

CED Q

result0_i__0 RTL_XOR

I0

I1 O

result0_i RTL_OR

I0[31:0]

I1[31:0] O[31:0] result_i RTL_MUX

S=3'b000 I0[31:0] S=3'b001 I1[31:0] S=3'b010 I2[31:0] S=3'b011 I3[31:0] S=3'b100 I4[31:0] S=3'b101 I5[31:0] S=default I6[31:0] O[31:0] S[2:0] =ZF_i RTL_EQO

I0[31:0]

I1[31:0] OF_i RTL_MUX

S=3'b000 I0

S=3'b001 I1

S=3'b010 I2

S=3'b011 I3

S=3'b100 I4

S=3'b101 I5

S=default I6 O

S[2:0]

CF_i RTL_MUX

S=3'b000 I0

S=3'b001 I1

S=3'b010 I2

S=3'b011 I3

S=3'b100 I4

S=3'b101 I5

S=default I6 O

S[2:0]

regs_reg[31]_i RTL_ROM

A[4:0] O

result_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0] pc_reg0_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0] pc_reg_reg[31:0] RTL_REG_ASYNC C

CLR D Q +

pc_plus4_i RTL_ADD

I0[31:0] V=B"100" I1[2:0] O[31:0] +

pc_branch_i RTL_ADD

I0[31:0]

I1[31:0] O[31:0] branch_taken_i RTL_AND

I0

I1 O

alu_b_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]